



### 6.1.5 Wrap-Up: Growth Mindset

"Many of life's failures are people who did not realize how close they were to success when they gave up." - Thomas Edison

1. If you start to feel like quitting, what will you do in that moment to persevere?
2. What would you want your teacher to say to you when they see you taking a risk, trying hard, or pushing through mistakes to pursue this goal?



## Unit 6, Lesson 2: Solving Compound Inequalities



### Benchmark

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.



### Learning Targets

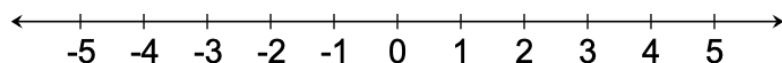
- ☐ I can solve compound one-variable inequalities.
- ☐ I can graph the solution of an inequality on a number line.
- ☐ I can interpret solutions to problems as viable or non-viable options.



### 6.2.1 Warm-Up: Bell Work

1. Solve and graph the inequality.

$$\frac{2}{3}(12m + 93) \leq 86 - (-m - 4)$$

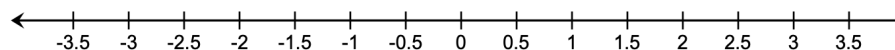


### 6.2.2 Exploration: Is It a Solution?

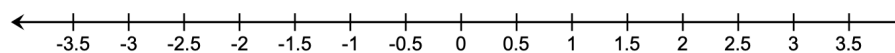
1. Two inequalities are shown in the table.
  - a. With your partner, discuss the similarities and differences of the two inequalities. Summarize your discussion.

| Inequality A                    | Inequality B                        |
|---------------------------------|-------------------------------------|
| $-17.5 < -\frac{1}{2}(8f + 11)$ | $-\frac{1}{2}(8f + 11) \leq 3f - 2$ |
|                                 |                                     |

- b. Solve each of the inequalities in the space provided in the table.
- c. Graph the solution set of Inequality A.



- d. Graph the solution set of Inequality B.



e. A list of values is shown.

|       |      |               |      |                |                |
|-------|------|---------------|------|----------------|----------------|
| -0.25 | 3    | $\frac{3}{4}$ | -84  | 1,000          | $-\frac{4}{5}$ |
| 2.87  | -1.4 | 0             | 18.6 | $-\frac{1}{2}$ |                |

With your partner, determine whether each value is in the solution set of Inequality A only, Inequality B only, or both. Record the values in the table below.

| Solution to Inequality A Only | Satisfies Both Inequalities | Solution to Inequality B Only |
|-------------------------------|-----------------------------|-------------------------------|
|                               |                             |                               |

f. Write two more numbers that are solutions to **both** Inequality A and Inequality B.

g. Describe the set of numbers that satisfies both Inequality A and Inequality B.



### 6.2.3 Guided Instruction: Compound Inequalities



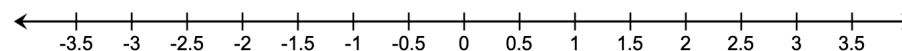
A **compound inequality** is a conjunction of two or more inequalities.

- The inequalities from the Exploration can be expressed as the compound inequality shown.

$$-17.5 < -\frac{1}{2}(8f + 11) \leq 3f - 2$$

The solution set of the compound inequality is the set of all values that satisfies both inequalities.

- Select the compound inequality that represents the solution set.
  - $-\frac{1}{2} < f \leq 3$
  - $-\frac{1}{2} \leq f < 3$
  - $3 < f \leq -\frac{1}{2}$
  - $3 \leq f < -\frac{1}{2}$
- Discuss with your partner what you think the graph of the solution set of this compound inequality would look like.
- The graph of the compound inequality is the **intersection** of the two graphs from the Exploration. Graph the solution set of the compound inequality.



The inequality from the previous question is an example of a **conjunctive** compound inequality.



In a **conjunction**, the solution set satisfies all of the inequalities and is the **intersection** of the solution sets. A **conjunction** often uses the word "and."

Compound inequalities can also be **disjunctive**.



In a **disjunction**, the solution set is any value that is in any of the individual solution sets. A **disjunction** often uses the word "or."

2. Consider the following example of a disjunctive compound inequality.

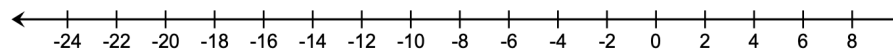
$$-3\left(1\frac{1}{2}m - 2\frac{3}{4}\right) < 22\frac{1}{4} - m \text{ or } \frac{75m-9}{12} \leq -132$$

- a. Solve each inequality.

$$-3\left(1\frac{1}{2}m - 2\frac{3}{4}\right) < 22\frac{1}{4} - m$$

$$\frac{75m-9}{12} \leq -132$$

- b. The solution set of the compound inequality is the set of all values that is a solution to either one of the inequalities. Graph the solution sets of **both** inequalities on the number line.



- c. Explain which of the following values is **not** a solution to the compound inequality.

-378

-5

19



### 6.2.4 Collaboration: Special Cases in Compound Inequalities

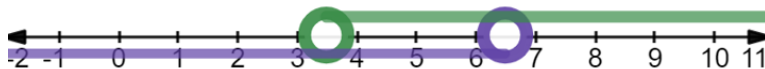
1. A disjunctive compound inequality is shown.

$$-22k + 5 < -14k - 23 \quad \text{or} \quad -4 < -3\left(k + \frac{1}{2}\right) + 17$$

- a. Solve each component of the compound inequality.

|                        |                                            |
|------------------------|--------------------------------------------|
| $-22k + 5 < -14k - 23$ | $-4 < -3\left(k + \frac{1}{2}\right) + 17$ |
|------------------------|--------------------------------------------|

- b. The graph of the solution set is shown.



What do you notice about the graph of the two solution sets?

- c. Luz says that 3.5 is not a solution to the compound inequality because the first inequality is strictly “greater than.” Explain why Luz is incorrect.
- d. Discuss with your partner which of the following sets describes the solution set of the compound inequality. Explain your choice.
- $\{k: k \in \mathbb{N}\}$
  - $\{k: k \in \mathbb{Q}\}$
  - $\{k: k \in \mathbb{R}\}$
  - $\{k: k \in \mathbb{Z}\}$

2. A conjunctive compound inequality is shown.

$$-1 < -z + \frac{3}{4}(z - 16) < \frac{1}{2}z + 12$$

- a. Rewrite the compound inequality as two single inequalities and solve each one.

|  |  |
|--|--|
|  |  |
|--|--|

- b. Explain whether any of the values shown satisfy **both** of the inequalities.

192.9

22

$-\frac{1}{3}$

- c. Describe the solution set of the compound inequality.



A compound inequality could have no solutions, a finite set of solutions, or a solution of all real numbers.



### 6.2.5 Guided Practice: Compound Inequalities in the Real World

1. Cladonia perforata is an endangered lichen found in the Atlantic Coastal Ridge, Lake Wales Ridge, Southwest Florida, and the North Gulf Coast of Florida. The habitat of this species is the white sand of the Florida scrub, already a rare and endangered type of ecosystem, and the lichen requires a specific spot within the habitat. To study the lichen after removing it from its ecosystem, botanists soak the lichen in water and then measure its surface area.



A botanist found that the lichen from Tiger Creek Preserve on the Lake Wales Ridge had surface areas that ranged between 431 square millimeters ( $\text{mm}^2$ ) and 693  $\text{mm}^2$ . The botanist uses the inequality shown, where  $w$  is the weight in grams (g), to determine the weight range of the lichen.

$$431 \leq 524w + 85.16 \leq 693$$

- a. Solve the inequality.
- b. Check your solution with your partner.

- c. Complete the statement.

The lichens in Tiger Creek Preserve have a weight of between \_\_\_\_\_ and \_\_\_\_\_ grams.

- d. Explain whether or not the *Cladonia perforata* in Tiger Creek Preserve could have a weight of 1 g.

- e. Explain whether or not the *Cladonia perforata* in Tiger Creek Preserve could have a weight of 0.6895 g.

- f. Explain why any solution to the mathematical inequality is a viable solution given the real-world context.



### 6.2.6 Your Turn!

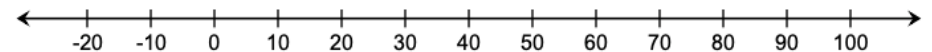
1. The Homeland Security and Emergency Management Agency has emergency procedures in place for extreme temperatures in Washington, D.C.

A temperature emergency is triggered if the temperature drops below 15° Fahrenheit (°F) or reaches 92°F. The compound inequality shown, where  $c$  is the temperature in degrees Celsius, can be used to determine the temperature in Celsius that would trigger a temperature emergency in Washington, D.C.

$$\frac{9}{5}c + 32 < 15 \quad \text{or} \quad \frac{9}{5}c + 32 \geq 92$$

- a. Solve the compound inequality.

- b. Graph the solution set of the compound inequality. Approximate values as necessary.



- c. Complete the statement.

A temperature emergency is declared in Washington, D.C. if the temperature is less than \_\_\_\_\_ °C or at least \_\_\_\_\_ °C.



### 6.2.7 Wrap-Up: Error Analysis

1. Simeon solved a compound inequality. His work is shown.

$$18 < -3y + 72 < \frac{1}{3}(168y - 92)$$

$$\begin{array}{lcl} & \swarrow & \searrow \\ 18 < -3y + 72 & & 18 < \frac{1}{3}(168y - 92) \\ -54 < -3y & & 18 < 56y - 46 \\ 18 > y & & 54 < 56y \\ & & \frac{54}{56} < y \\ & & \frac{27}{28} < y \end{array}$$

Simeon stated the solution set is  $\frac{27}{28} < y < 18$ .

Explain the mistake(s) Simeon made.

## Unit 6, Lesson 3: Writing and Solving Inequalities



### Benchmark

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.



### Learning Targets

- ☐ I can write and solve a two-step, one-variable inequality.
- ☐ I can write and solve a multi-step, one-variable inequality.
- ☐ I can write and solve a compound one-variable inequality.
- ☐ I can interpret solutions to problems as viable or non-viable options.